Question #1 of 11

Question ID: 1472237

Under which of these conditions is a machine learning model said to be underfit?

A) The model identifies spurious relationships.

B) The model treats true parameters as noise.

C) The input data are not labelled.

Question #2 of 11

Question ID: 1472238

When evaluating the fit of a machine learning algorithm, it is *most* accurate to state that:

A) accuracy is the ratio of correctly predicted positive classes to all predicted positive classes.

B) recall is the ratio of correctly predicted positive classes to all actual positive classes.

C) precision is the percentage of correctly predicted classes out of total predictions.

Question #3 of 11

Question ID: 1472234

Big data is *most* likely to suffer from low:

A) velocity.

B) variety.

C) veracity.

Question #4 of 11

Question ID: 1472232

An executive describes her company's "low latency, multiple terabyte" requirements for managing Big Data. To which characteristics of Big Data is the executive referring?

A) Volume and velocity.

B) Velocity and variety.

C) Volume and variety.

Question #5 of 11

Question ID: 1472236

In big data projects, data exploration is *least* likely to encompass:

A) feature selection.

B) feature engineering.

C) feature design.

Question #6 of 11

Question ID: 1472235

The process of splitting a given text into separate words is *best* characterized as:

A) bag-of-words.

B) stemming.

C) tokenization.

Question #7 of 11

Question ID: 1472233

Which of the following uses of data is *most accurately* described as curation?

A) An analyst adjusts daily stock index data from two countries for their different market holidays.

B) An investor creates a word cloud from financial analysts' recent research reports about a company.

C) A data technician accesses an offsite archive to retrieve data that has been stored there.

Freja Karlsson is a bond analyst with Storbank AB. Over the past several months, Karlsson has been working to develop her own machine learning (ML) model that she plans to use to predict default of the various bonds that she covers. The inputs to the model are various pieces of financial data that Karlsson has compiled from multiple sources.

After Karlsson has constructed the model using her knowledge of appropriate variables, Karlsson runs the model on the training set. Each firm's bonds are classified as predicted- to-default or predicted-not-to-default. When Karlsson's model predicts that a bond will default and the bond actually defaults, Karlsson considers this to be a true positive. Karlsson then evaluates the performance of her model using error analysis. The confusion matrix that results is shown in Exhibit 1.

N = 474		**Actual Bond Status**	
		Bond Default	**No Default**
Model Prediction	Bond Default	307	31
	No Default	23	113

Question #8 - 11 of 11

Question ID: 1472240

Based on Exhibit 1, Karlsson's model's precision is *closest* to:

A) 71%.

B) 81%.

C) 91%.

Question #9 - 11 of 11

Question ID: 1472241

Karlsson is especially concerned about the possibility that her model may indicate that a bond will not default, but then the bond actually defaults. Karlsson decides to use the model's recall to evaluate this possibility. Based on the data in Exhibit 1, the model's recall is *closest* to:

A) 93%.

B) 83%.

C) 73%.

Question #10 - 11 of 11

Question ID: 1472242

Karlsson would like to gain a sense of her model's overall performance. In her research, Karlsson learns about the F1 score, which she hopes will provide a useful measure. Based on Exhibit 1, Karlsson's model's F1 score is *closest* to:

A) 82%.

B) 92%.

C) 72%.

Question #11 - 11 of 11

Question ID: 1472243

Karlsson also learns of the model measure of accuracy. Based on Exhibit 1, Karlsson's model's accuracy metric is *closest* to:

A) 89%.

B) 69%.

C) 79%.